QA ENGINEER I / TECHNICAL ASSESSMENT

Currency Converter

Requirements under test

Your test cases must cover every requirement below. Each test case should trace back to at least one requirement ID.

01 Input validation

R1	Throw IllegalArgumentException if amount <= 0
R2	Throw IllegalArgumentException if the source or target currency code is null or empty

02 Conversion flow

R3	Input the amount into the amount field
R4	Select the source currency from the dropdown
R5	Select the target currency from the dropdown
R6	Wait for the result container to become visible
R7	Extract the converted amount and the exchange rate
R8	Format the input amount to 2 decimal places using String.format("%.2f", amount)
R9	Count decimal places in both the formatted input amount string and the result string

03 Results validation

R10	Accuracy: Math.abs(conversionResult - amount * rate) <= 0.01
R11	Decimal precision: the result has 2 or more decimal places
R12	Dynamic update: modify the input amount, confirm the result updates by comparing before/after values, then reset the GUI to the original amount to ensure a clean state

04 Returned ConversionResult - all fields populated

R13	`inputAmount`	The exact amount entered into the amount field (e.g. 100.0)
R14	`sourceCurrency`	The code selected in the source dropdown (e.g. "USD")
R15	`targetCurrency`	The code selected in the target dropdown (e.g. "EUR")
R16	`conversionResult`	The converted amount extracted from the result field in the UI (e.g. 92.35)
R17	`inputDecimalPlaces`	Decimal places in the formatted input string (e.g. 2 for "100.00")
R18	`resultDecimalPlaces`	Decimal places in the result string displayed in the UI (e.g. 2 for "92.35")
R19	`conversionAccurate`	true only if Math.abs(conversionResult - amount * rate) <= 0.01
R20	`decimalPrecisionMaintained`	true only if the result has 2 or more decimal places
R21	`resultUpdatedOnChange`	true only if the result updates when the input changes
R22	`conversionComplete`	true only if all of the above conditions are met